

VLSI-Based Intelligent Attendance Register Using Real-Time Face Recognition

PROGRAM CODED IN PROJECT:

Vivado source code:

```
`timescale 1ns/1ps

module attendance #(parameter N=8, WIDTH=8) ( // N = number of students, WIDTH = ID
width
    input  wire clk,
    input  wire rst,
    output reg [N-1:0] present_bits // 1 = present, 0 = absent
);

    // Memory for enrolled IDs (template) and detected IDs (features)
    reg [WIDTH-1:0] template [0:N-1]; // enrolled students
    reg [WIDTH-1:0] features [0:N-1]; // detected faces

    integer i, j;

    // Initialize attendance register
    always @(posedge clk or posedge rst) begin
        if (rst) begin
            present_bits <= 0;
        end else begin
            // Compare features with template
```

```

    for (i = 0; i < N; i = i + 1) begin
        for (j = 0; j < N; j = j + 1) begin
            if (features[i] == template[j]) begin
                present_bits[j] <= 1; // mark present
            end
        end
    end
end
end
end
end

```

```

// Load template and features from HEX files

```

```

initial begin

```

```

    $readmemh("template.hex", template);

```

```

    $readmemh("features.hex", features);

```

```

end

```

```

endmodule

```

vivado test bench code:

```

`timescale 1ns/1ps

```

```

module attendance #(parameter N=8, WIDTH=8) ( // N = number of students, WIDTH = ID
width

```

```

    input  wire clk,

```

```

    input  wire rst,

```

```

    output reg [N-1:0] present_bits // 1 = present, 0 = absent

```

```

);

```

```
// Memory for enrolled IDs (template) and detected IDs (features)
```

```
reg [WIDTH-1:0] template [0:N-1]; // enrolled students
```

```
reg [WIDTH-1:0] features [0:N-1]; // detected faces
```

```
integer i, j;
```

```
// Initialize attendance register
```

```
always @(posedge clk or posedge rst) begin
```

```
    if (rst) begin
```

```
        present_bits <= 0;
```

```
    end else begin
```

```
        // Compare features with template
```

```
        for (i = 0; i < N; i = i + 1) begin
```

```
            for (j = 0; j < N; j = j + 1) begin
```

```
                if (features[i] == template[j]) begin
```

```
                    present_bits[j] <= 1; // mark present
```

```
                end
```

```
            end
```

```
        end
```

```
    end
```

```
end
```

```
// Load template and features from HEX files
```

```
initial begin
```

```
    $readmemh("template.hex", template);
```

```
    $readmemh("features.hex", features);
```

```
end
```

```
endmodule
```

app.py in visual studio:

```
import cv2
import face_recognition
from datetime import datetime

students = {
    "621323106102 - Shalini K": "shalini.jpg",
    "621323106304 - Subashri A": "subasri.jpg",
    "621323106121 - Vigneshwari P": "vigneshwari.jpg"
}

known_encodings = []
known_names = []
for name, img_file in students.items():
    img = face_recognition.load_image_file(img_file)
    enc = face_recognition.face_encodings(img)[0]
    known_encodings.append(enc)
    known_names.append(name)

cap = cv2.VideoCapture(0)
detected_ids = []
```

```
print("Press 'q' to stop camera and generate features.hex")
```

```
while True:
```

```
    ret, frame = cap.read()
```

```
    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
```

```
    face_locations = face_recognition.face_locations(rgb_frame)
```

```
    face_encs = face_recognition.face_encodings(rgb_frame, face_locations)
```

```
    for enc in face_encs:
```

```
        matches = face_recognition.compare_faces(known_encodings, enc)
```

```
        if True in matches:
```

```
            idx = matches.index(True)
```

```
            name = known_names[idx]
```

```
            student_id = idx + 1
```

```
            detected_ids.append(student_id)
```

```
            cv2.putText(frame, name, (50,50), cv2.FONT_HERSHEY_SIMPLEX, 1, (0,255,0), 2)
```

```
    cv2.imshow("Camera", frame)
```

```
    if cv2.waitKey(1) & 0xFF == ord('q'):
```

```
        break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

```

detected_ids = list(set(detected_ids))

with open(r"D:\Attendance\attendance\attendance.sim\sim_1\behav\xsim\features.hex", "w") as f:

    for sid in detected_ids:

        f.write(f'{sid:02X}\n')

print("features.hex created with:", detected_ids)

```

main.py in visual studio:

```

import pandas as pd

from datetime import datetime

id_to_name = {1:"621323106102 - Shalini K", 2:"621323106304 - Subashri A",
3:"621323106121 - Vigneshwari P"}

with open(r"D:\Attendance\attendance\attendance.sim\sim_1\behav\xsim\present.hex") as f:

    print("READED")

    bitmask = int(f.read().strip(), 2)

attendance = []

for sid, name in id_to_name.items():

    if (bitmask >> (sid-1)) & 1:

        now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

        attendance.append([name, now])

df = pd.DataFrame(attendance, columns=["Name", "Time"])

df.to_csv("attendance.csv", index=False)

```

```
print(df)
```